

SIDDHARTH KUMAR

Full Stack Developer — AI/ML Engineer — Cybersecurity Enthusiast

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PROFESSIONAL SUMMARY

Highly motivated Computer Science Engineering student with a strong foundation in Full Stack Development, Artificial Intelligence/Machine Learning, and Cybersecurity. Demonstrated expertise in building scalable web applications, intelligent systems, and security-focused solutions. Proven track record of leading technical teams to success in competitive hackathons and delivering impactful real-world projects. Experienced in end-to-end software development lifecycle with proficiency in modern frameworks, cloud technologies, and AI/ML tools. Passionate about leveraging technology to solve complex problems and create meaningful user experiences.

EDUCATION

Bachelor of Technology in Computer Science Engineering

August 2023 - May 2027

Sikkim Manipal Institute of Technology, Sikkim, India

Minor: Cybersecurity — **Current Status:** 6th Semester (3rd Year)

Relevant Coursework:

- Data Structures and Algorithms, Database Management Systems, Computer Networks and Security
- Artificial Intelligence and Machine Learning, Software Engineering and Development
- Operating Systems, Web Technologies, Cloud Computing

Higher Secondary Education (12th Grade) - Science (PCM)

2020 - 2022

Percentage: 70% — JEE Mains: 91 percentile — JEE Advanced: Qualified — WBJEE Rank: Under 8000

PROFESSIONAL EXPERIENCE

Summer Intern - Department of Information Technology

June 2025 - August 2025

North Bihar Power Distribution Company Limited (NBPDC) — Bihar, India

- Developed comprehensive end-to-end electricity theft detection system using machine learning and anomaly detection algorithms to identify irregular consumption patterns
- Built interactive Streamlit dashboard with advanced data visualization capabilities using Pandas and Plotly for real-time consumption pattern analysis
- Implemented automated flagging system that reduced manual inspection time by 60%, significantly improving operational efficiency and resource allocation
- Processed and analyzed over 50,000 customer consumption records using Python, NumPy, and SQLite database management
- Designed data preprocessing pipelines for cleaning, normalizing, and transforming raw meter reading data
- Presented technical findings, project methodology, and business impact to senior management and received official commendation for innovation
- Collaborated with cross-functional teams including field engineers, data analysts, and management personnel
- **Technologies:** Python, Streamlit, Pandas, Plotly, NumPy, SQLite, Scikit-learn, Data Analysis

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, C++, HTML5, CSS3

Frontend Development: React.js, Next.js, TypeScript, Tailwind CSS, Material-UI, shadcn/ui, Vite, Redux, React Query, Responsive Design

Backend Development: Python (FastAPI, Flask, Django), Node.js, Express.js, RESTful API Design, Microservices Architecture

Database Technologies: MongoDB, PostgreSQL, SQLite, Firebase, Database Design, Query Optimization

AI/ML & Data Science: TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Hugging Face Transformers, spaCy, LangChain, OpenCV, MediaPipe, NLP, Computer Vision, ONNX, TensorFlow.js

Browser Extensions & APIs: Chrome Extensions API, WebExtensions, Google WebRisk API, Wolfram API

AR/VR Technologies: Unity, ARCore, ARKit, WebXR, 360° Media, Immersive Experiences

DevOps & Cloud: Git, GitHub, Docker, Linux/Unix, CI/CD, AWS Services, Cloud Deployment

Web Security & Cybersecurity: Phishing Detection, Heuristic Analysis, Threat Intelligence, Web Security Best Practices, OWASP, Secure Coding

Additional Tools: Streamlit, Postman, VS Code, Jupyter Notebook, Figma, Agile/Scrum

KEY PROJECTS

PhishDetectAI - AI-Powered Browser Security Extension

Role: Full-Stack Developer — AI & Cybersecurity Engineer

Project Overview: Privacy-first, AI-driven browser extension designed to protect users from phishing attacks, malicious links, and unsafe web resources in real-time.

- Built AI-powered Chrome extension for real-time phishing detection using heuristic analysis and Google WebRisk API with sub-100ms latency
- Designed privacy-first, local-execution security model combining rule-based analysis and ML risk-scoring for URLs
- Implemented dynamic ad and tracker blocking system capable of filtering obfuscated scripts on high-risk websites
- Developed FastAPI backend optimized for scalable threat verification with email and document scanning capabilities
- Integrated Wolfram-powered Q&A assistant to explain security warnings in simple, user-friendly language
- Conducted live demonstrations showing effective phishing detection and intrusive ad removal
- **Impact:** Successfully demonstrated production-ready cybersecurity solution with minimal performance overhead
- **Technologies:** Python, FastAPI, JavaScript, Chrome Extensions API, Google WebRisk API, Wolfram API, ML

NeuroDoc - AI-Powered Document Intelligence Platform

Role: Full-Stack Developer — AI/ML Engineer

Project Overview: End-to-end AI-driven document intelligence platform for automatic summarization, highlighting, validation, and interaction with large text documents.

- Developed comprehensive document intelligence platform with extractive and abstractive summarization using transformer models (BART, T5)
- Built interactive PDF chat system with Q&A interface enabling natural language queries on uploaded documents
- Implemented AI-powered podcast generation feature converting documents to audio using text-to-speech synthesis
- Designed client-side ML execution using ONNX and TensorFlow.js for privacy-first processing of sensitive documents
- Created smart highlighting engine with keyword extraction, sentence scoring, and attention-based techniques
- Implemented automated content validation and bias detection to improve summary reliability
- Built automated quiz generation system from document content to enhance learning and retention
- **Impact:** Achieved 70% reduction in document review time and 85% improvement in information extraction efficiency
- **Technologies:** React 18, TypeScript, Vite, FastAPI, Hugging Face Transformers, spaCy, TensorFlow.js, ONNX, FAISS, Tailwind CSS, shadcn/ui

MyBudgetAI - Smart Financial Assistant

Role: Full-Stack Developer — Data Analyst

Project Overview: Intelligent personal finance analysis tool that processes real bank transaction data to provide spending insights and recommendations.

- Built personal finance analysis web application processing real bank transaction CSV data with automated data cleaning and validation
- Implemented dynamic expense classification and filtering system to isolate spending transactions
- Developed interactive data visualizations including category-wise spending bar charts and monthly trend analysis
- Engineered rule-based financial recommendation system generating actionable saving insights based on spending behavior
- Enabled users to identify high-spending categories with suggested reduction percentages for budget optimization
- Optimized data processing using Pandas aggregation and resampling techniques for scalable financial analysis
- **Technologies:** Python, Streamlit, Pandas, Matplotlib, Seaborn, NumPy, Data Analysis

Monastery360 - Digital Heritage & Smart Tourism Platform

Role: AR/VR Developer — Full-Stack Developer — Team: HeritEdge

Project Overview: Digital heritage platform designed to preserve and showcase monasteries of Sikkim through immersive AR/VR technology and smart tourism integration.

- Designed immersive 360° AR/VR experience for digital preservation of 200+ monasteries in Sikkim with interactive cultural narration
- Implemented offline-first architecture enabling downloadable monastery information packs (360° views, audio guides, maps) for low-connectivity regions
- Integrated smart tourism features including interactive maps, travel route guidance, weather and road condition

awareness

- Developed QR-based AR interactions at monastery sites to deliver contextual information while respecting cultural restrictions
- Designed community-centric model involving local guides, transport providers, and artisans for economic empowerment
- Built cultural calendar system highlighting festivals, rituals, and events to encourage time-specific tourism
- **Impact:** Created scalable framework for digital cultural preservation and sustainable tourism
- **Technologies:** React, Flutter, Unity, ARCore/ARKit, WebXR, FastAPI, Firebase, PostgreSQL, Google Maps API

Indian Sign Language Translator

Role: Computer Vision Engineer — ML Developer

Project Overview: Accessibility-focused real-time computer vision application to translate Indian Sign Language to text and speech.

- Developed real-time computer vision system recognizing 26 ISL gestures with 87% accuracy at 25 FPS
- Implemented hand tracking and landmark detection using MediaPipe framework
- Built gesture classification model using TensorFlow with custom dataset collection and augmentation
- Designed web interface for seamless user interaction and real-time translation display
- **Technologies:** Python, OpenCV, MediaPipe, TensorFlow, Keras, React, WebRTC

Additional Projects:

- **Electricity Theft Detection System (NBPDCI):** ML system for consumption pattern analysis and anomaly detection
- **Instagram Unfollow Checker:** Chrome extension developed in 2 hours to identify non-followers

LEADERSHIP POSITIONS & EXTRACURRICULAR ACTIVITIES

Chief Website Administrator

January 2026 - Present

SMIT Model United Nations (SMIT MUN)

- Lead administrator for official conference website, managing delegate registrations and committee allocations
- Optimized server performance and database queries to handle concurrent traffic spikes during registration phases
- Implemented secure payment gateway integrations for conference fee processing
- Developed automated email notification system for delegates and committee members
- Managed website security, uptime monitoring, and bug fixes

AI/ML Co-Lead - Encoders (Coding Club), SMIT

- Co-leading AI/ML initiatives and technical workshops for the college coding club
- Conducted comprehensive C++ programming workshop for 80+ first and second-year students
- Mentoring students in AI/ML projects, competitive programming, and hackathon preparation
- Organizing coding competitions, study groups, and collaborative learning sessions

Event Management, Aesthetics & Production Lead - CINEMATES

Acting & Cinematography Club, SMIT

- Leading event planning, execution, and production for college cultural and cinematography events
- Managing aesthetics, stage design, and creative direction for performances

Former Positions:

- **PR Lead Artist** - SMIT: Led public relations and promotional campaigns for college events
- **PR Co-Lead** - Codhers (Coding Community): Co-led PR efforts for coding community
- **Member** - Enactus, Institution's Innovation Council (IIC), Innovison, Internshala Student Partner

HACKATHONS & COMPETITIONS

- **Tech Adrishta 2025** - Runner-Up, College Annual Tech Fest — Team Lead
- **InnoFusion 2.0** - Top 5 Finalist (from 3,300+ participants to Top 25 finalists), UEM Kolkata, 2024 — Team Lead
- **Devopia 2.0** - Finalist, Dell Technologies Partner Hackathon, 2024 — Team Lead
- **Smart India Hackathon (SIH)** - Qualified internal rounds in 2024 & 2025 — Team Lead
- **EnCode 2026** - Participated, IIT Guwahati — Team Lead
- **InnovWar Hackathon 2025** - 24-hour competitive coding challenge
- **World's Largest Hackathon by Bolt** - Participated, Devpost, 2024
- **Hawk hacks** - Participated, Devpost, 2024

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- DevOps and AI on AWS: Upgrading Apps with Generative AI - Coursera/AWS — Grade: 100%
- PostgreSQL Fundamentals - Coursera/AWS — Grade: 100%
- Deloitte Virtual Internship in Cybersecurity - Forage
- Full Stack Developer Bootcamp - GeeksforGeeks
- Complete TensorFlow Course - GeeksforGeeks
- Google AI Essentials - Coursera/Google
- Foundations of Cybersecurity - Coursera/Google
- Generative AI for Software Development - Coursera
- Introduction to DevOps - Coursera/IBM
- Foundations of Augmented Reality - Coursera/Meta
- GFG 160 - 160 Days of Problem Solving - GeeksforGeeks
- Free Python Course with Certificate - GeeksforGeeks

ADDITIONAL INFORMATION

Languages: English (Fluent), Hindi (Fluent), Bengali (Professional)

Core Competencies:

- End-to-End Full Stack Development, AI/ML Model Development and Deployment
- Browser Extension Development, Data Analysis and Visualization
- Team Leadership and Project Management, Problem-Solving and Critical Thinking
- Rapid Prototyping and Agile Development, Technical Documentation and Presentation

Interests: Artificial Intelligence, Cybersecurity, Web Development, Data Science, AR/VR Technologies, Open Source Contribution, Competitive Programming, Tech Communities

REFERENCES

Available upon request